ARLADI PUZZLES

Space Quest

Brain Adventurers

AGES 9–11 • SET 03

16 fresh picture-puzzle activities



Reason • Plan • Discover

Big ideas start here.

Work step by step. Check that your answer follows every clue. Sketch or write working when it helps.



This book belongs to

_____

1 LOOK closely

Notice the pictures and the clues.

2 TRY a plan

Point, draw, count, or sketch to help your thinking.

3 CHECK and smile

A new try is part of learning. There is no timer.

For grown-ups

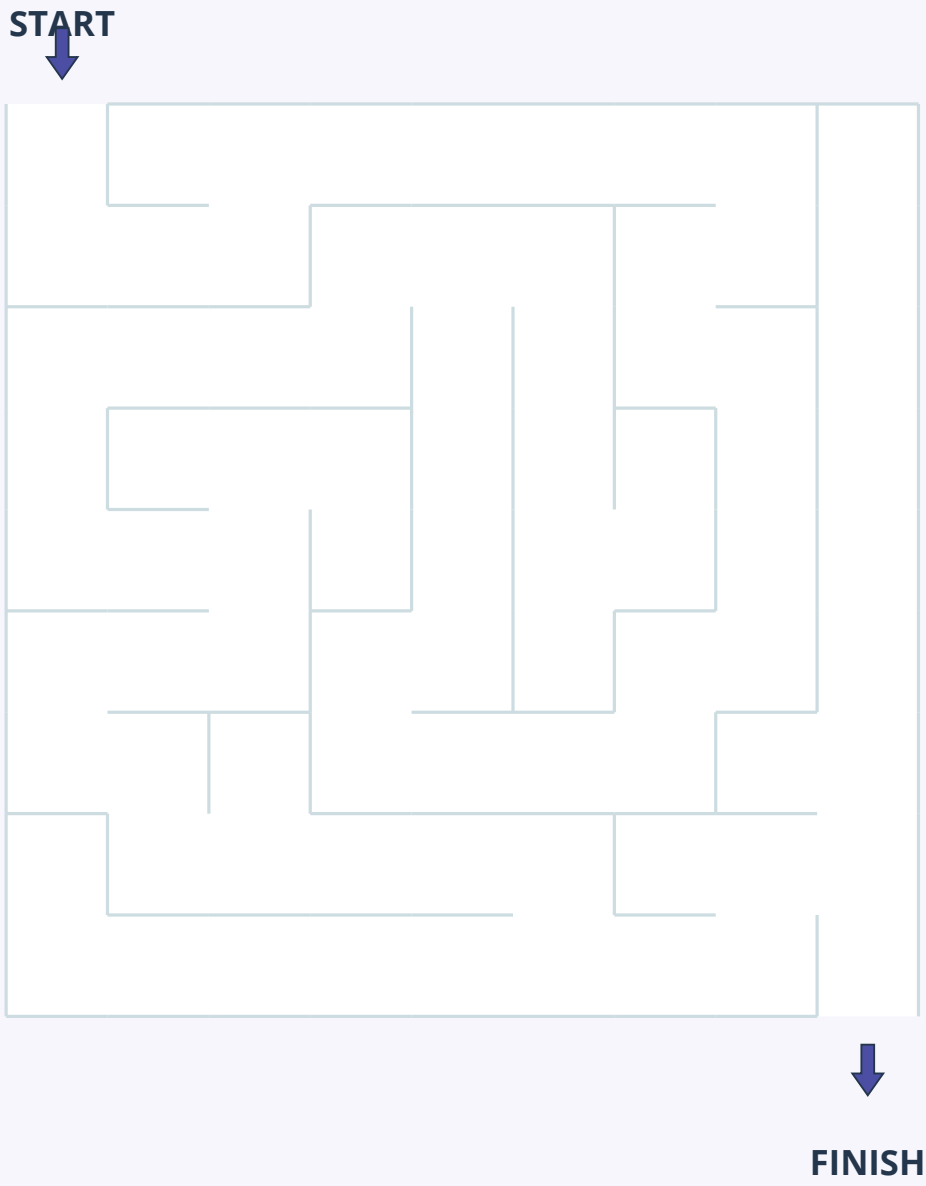
Combine visual reasoning with number patterns, spatial challenges, data, and multi-step logic.

Each activity has a matching answer card on pages 19–22. Drawing and memory activities include guidance for checking together. Let your child choose a comfortable pace.

PRINT: A4 portrait in color. For Letter paper, choose Fit.

The winding way

Find the only route from START to FINISH. Stay in the passages and never cross a wall.



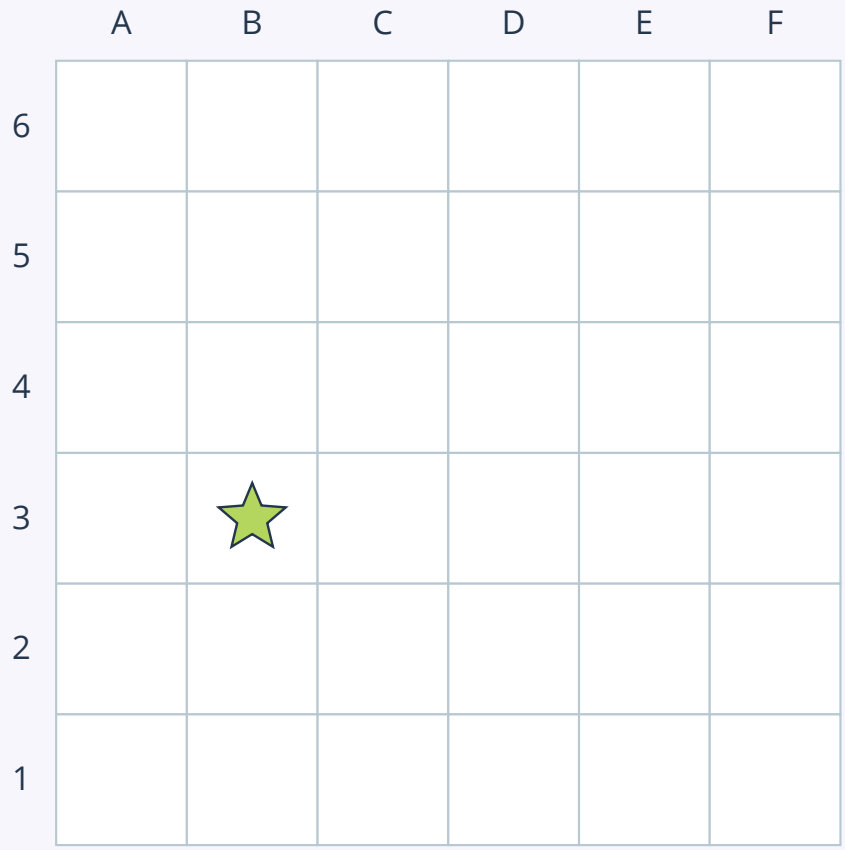
Coordinates in motion

Start at the star. Follow the arrows in order. One step moves to the next square. Name the finish coordinate.

START: B3

1 → 2 | 2 ← 3 | 3 ↓ 2 | 4 → 3

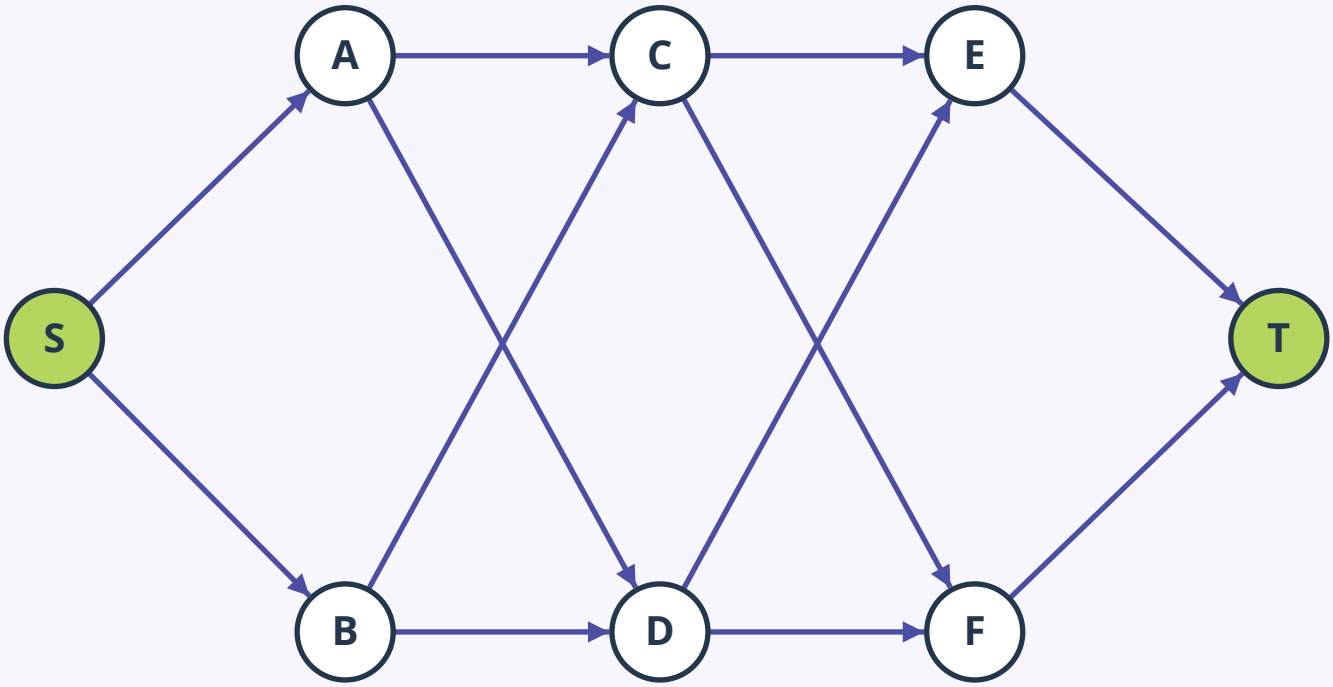
5 ↑ 2 | 6 ← 1 | 7 → 1



Finish coordinate: _____

How many different paths?

Count all routes from S to T. Follow arrow directions. Change paths only at circles, even where lines cross.



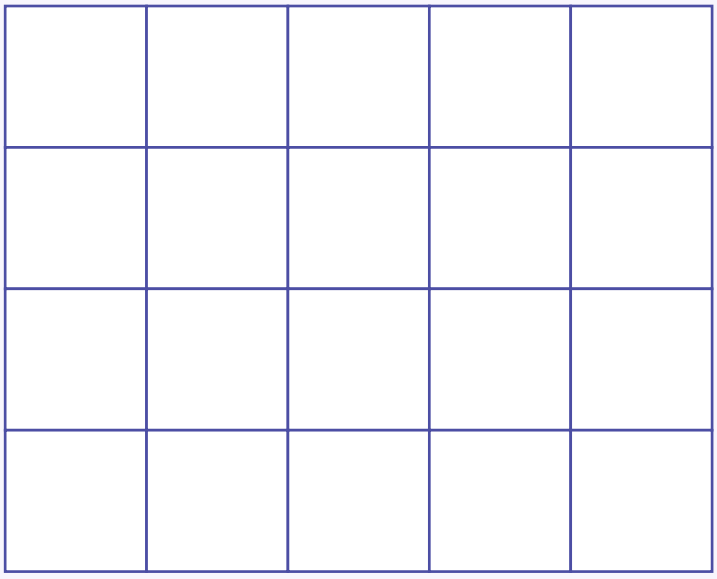
Write routes here, such as S-A-C-E-T:

Total routes:

Share the mosaic

There are 20 tiles. 4/10 get dots and 3/10 get stripes. No tile gets both. How many stay blank?

Draw dots and stripes. Keep the groups separate.

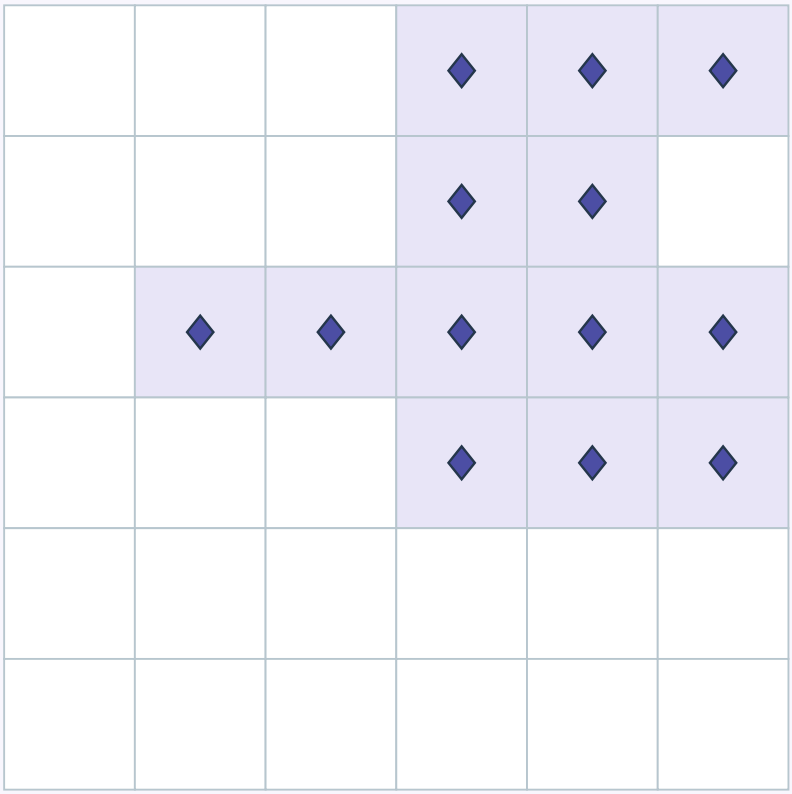


You may mark any tiles to help you count.

Blank tiles:

Walk around the edge

Each small square has side length 1 unit. Find the perimeter around the shaded shape. Shared edges inside do not count.



1 unit

Perimeter: _____ units

The shifted picture code

Each picture has a number. Add 3 to that number, then change it to a letter using A=1, B=2 and so on.

PICTURE KEY



Decode these pictures from left to right:

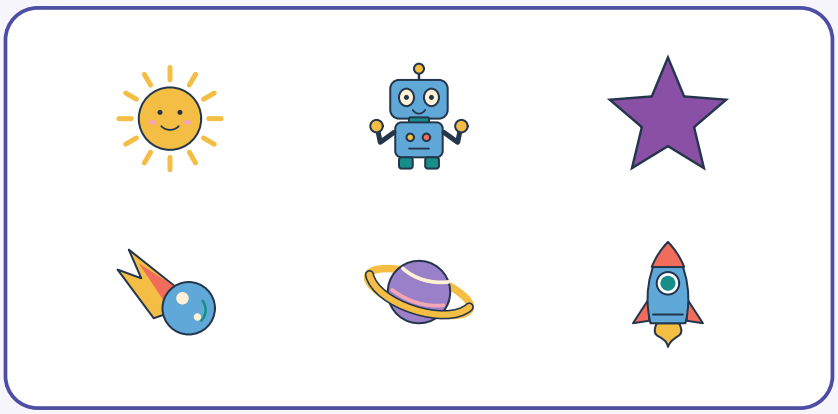


A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

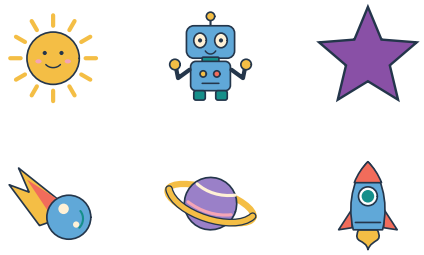
Remember six places

Study the top box. Cover it with paper, then choose the box with exactly the same six pictures in the same positions.

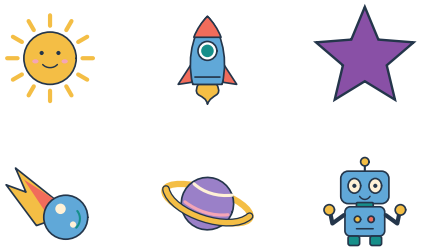


COVER THE TOP BOX

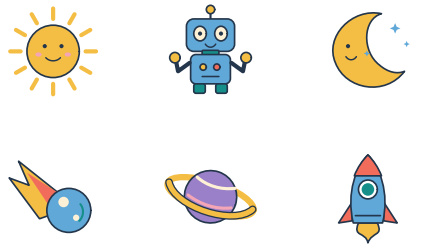
A



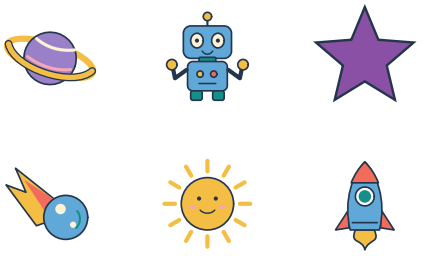
B



C



D



Read the picture votes

Each small star means 2 votes. How many more votes did rocket and moon get together than planet?

★ = 2 votes



★ ★ ★ ★ ★

rocket _____



★ ★ ★ ★ ★

moon _____



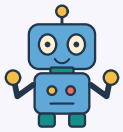
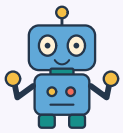
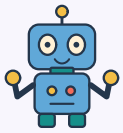
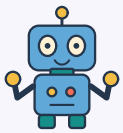
★ ★ ★

planet _____

How many more votes? ☐

Crack the number machine

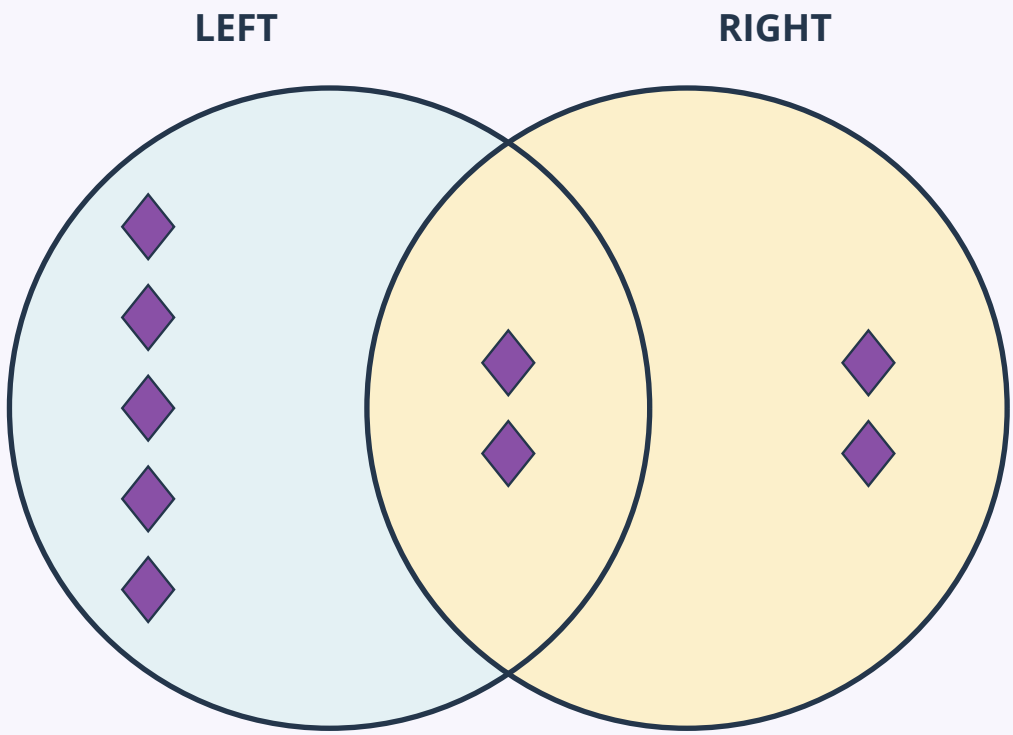
The rule is: multiply by a whole number from 2 to 5, then add a whole number from 1 to 9. The same rule fits every row.

INPUT		OUTPUT
4	→  →	25
7	→  →	40
10	→  →	55
13	→  →	

The missing output is:

Overlap point hunt

Each diamond inside exactly ONE circle earns 2 points. Each diamond inside BOTH earns 4 points. Outside diamonds earn 0. Find the total.



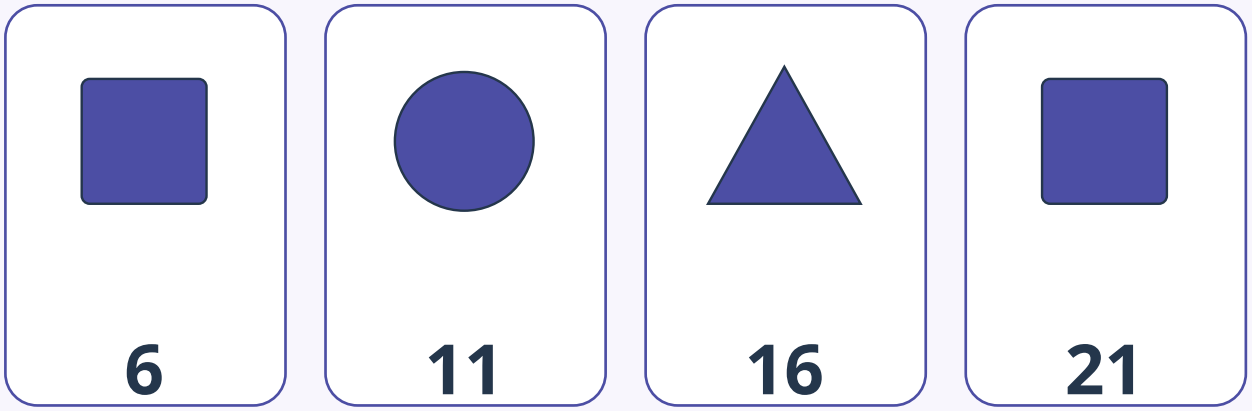


Outside

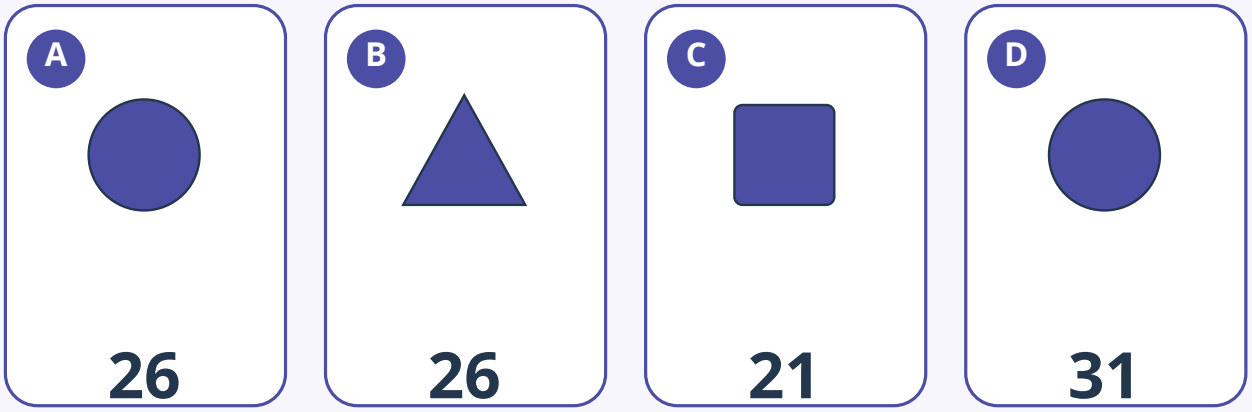
Total score:

Two rules at once

Find the next card. Follow both the shape rule and the number rule.

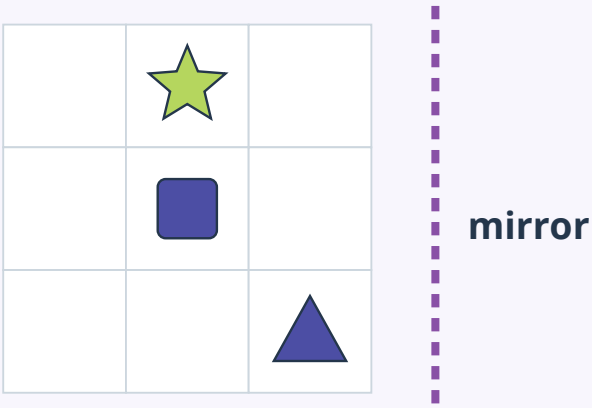


Which card comes next?

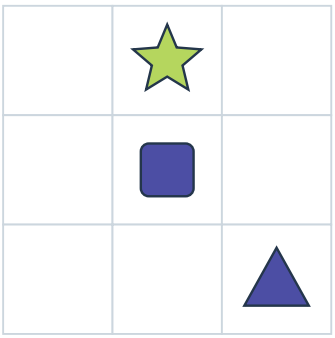


Move the whole group

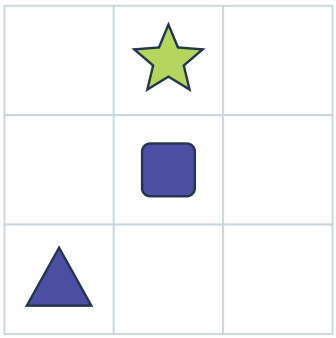
Reflect the WHOLE group in an upright mirror. Which option is the result?



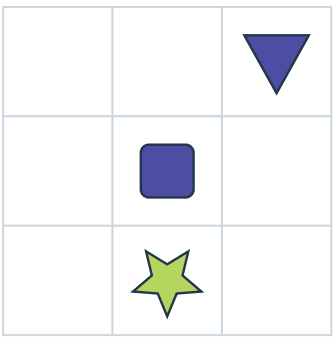
A



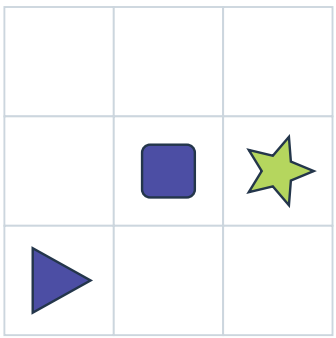
B



C



D



Picture equation quest

The same picture always has the same value. Solve the clues, then the final sum.

 +  = 10

 +  = 14

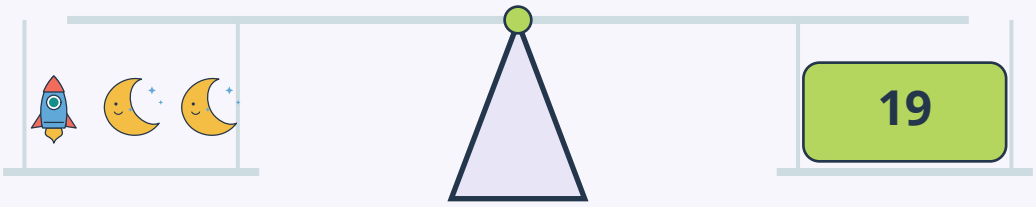
 +  = 16

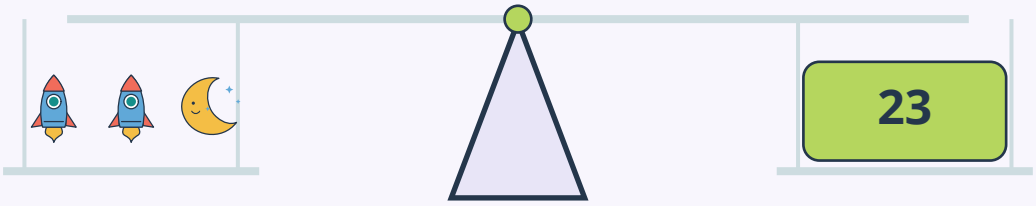
Now find:

 +  +  = ☐

Balanced treasures

Both scales balance. The yellow weights are in grams. Find the combined mass of one of each picture.





One of each:  +  = ☐ g

Four by four Sudoku

Write 1, 2, 3 and 4 once in every row, every column and every bold 2-by-2 box.

2	4		
1	3	4	
		1	

Check all three: rows, columns, small boxes.

Four friends, four finds

Each friend chose one different object. Use the clues and tick one square in each row and column.

Object order: left to right



rocket moon planet robot

1. Ira chose the object immediately left of Lina's.
2. Kai and Ira chose neighboring objects.
3. Kai and Jia chose neighboring objects.



Check your discoveries

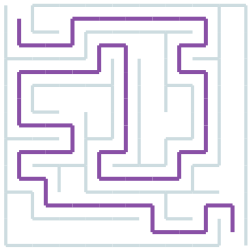
Puzzles 1–4 • Use the page number to find each puzzle.

01 • PAGE 3

The winding way

The route is shown in color.

The maze has a single route between its entrance and exit.

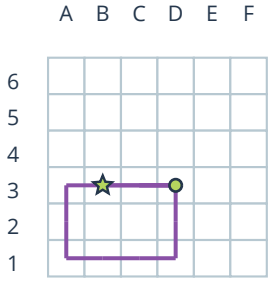


02 • PAGE 4

Coordinates in motion

D3

Read the column letter first and then the row number.



03 • PAGE 5

How many different paths?

8 routes

S-A-C-E-T; S-A-C-F-T; S-A-D-E-T; S-A-D-F-T; S-B-C-E-T; S-B-C-F-T; S-B-D-E-T; S-B-D-F-T

04 • PAGE 6

Share the mosaic

6 blank tiles

Dotted: 8. Striped: 6. Subtract both from 20. Any nonoverlapping positions work.

Check your discoveries

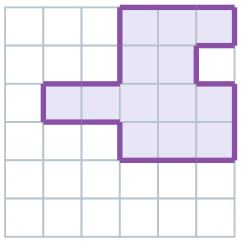
Puzzles 5–8 • Use the page number to find each puzzle.

05 • PAGE 7

Walk around the edge

20 units

Count each exposed square edge once. Do not count edges between two shaded squares.



06 • PAGE 8

The shifted picture code

NORTH

Add 3 to each key number, then use the alphabet strip. Read the five decoded letters in order.

07 • PAGE 9

Remember six places

A

Top row: sun, robot, star. Bottom row: comet, planet, rocket.

08 • PAGE 10

Read the picture votes

14 votes

Totals are 10, 10 and 6. Calculate 10 + 10 - 6.

Check your discoveries

Puzzles 9–12 • Use the page number to find each puzzle.

09 • PAGE 11

Crack the number machine

70

Multiply by 5, then add 5. So 13 x 5 + 5 = 70.

10 • PAGE 12

Overlap point hunt

22 points

7 diamonds are in exactly one circle; 2 are in both. 7 x 2 + 2 x 4 = 22. Ignore the 1 outside.

11 • PAGE 13

Two rules at once

A: circle, 26

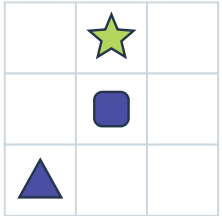
The shapes repeat in groups of three. Add 5 each time to the number.

12 • PAGE 14

Move the whole group

B

Every item moves with the group. Keep all three shapes in their correct transformed positions.



Check your discoveries

Puzzles 13–16 • Use the page number to find each puzzle.

13 • PAGE 15

Picture equation quest

19

rocket = 5, moon = 9, planet = 7. Use those values in the last sum.

14 • PAGE 16

Balanced treasures

14 g

The first picture is 9 g and the second is 5 g. They satisfy both balanced scales.

15 • PAGE 17

Four by four Sudoku

Completed grid shown.

There is exactly one completion that meets all three rules.

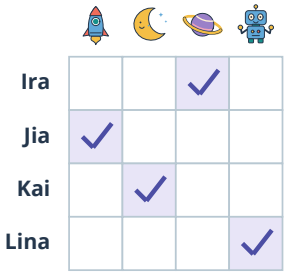
2	4	3	1
3	1	2	4
1	3	4	2
4	2	1	3

16 • PAGE 18

Four friends, four finds

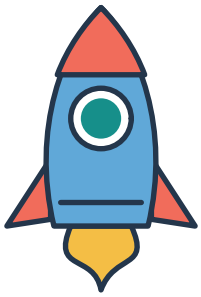
Matching grid shown.

Each friend chose exactly one object. Check every clue against the ticked pairs.



CERTIFICATE OF

Puzzle Power!



Awarded to

_____

for looking closely, thinking carefully,
and trying new ways to solve a puzzle.

Your ideas matter.

_____ _____

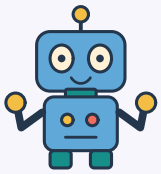
DATE CHEERED ON BY

KEEP YOUR CURIOSITY GROWING

Space Quest

Brain Adventurers





A book of fresh puzzles for curious minds.

Try a new page. Share a discovery.

Come back to a favorite challenge.

Five themed sets. Three age groups. Lots to discover.

ARLADI PUZZLES